



Towards fiscal sustainable practices in Mexico: The quality of municipal tax regulation and tiered tariffs improve income from solid waste management services

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ABSTRACT

Solid waste management (SWM) is a fundamental responsibility of local governments, essential for public health and environmental sustainability. Globally, fiscal practices, such as tax regulation and tariff structures, are critical for ensuring effective SWM services and achieving sustainable development goals. In Mexico, municipalities face financial constraints and heavily depend on subsidies rather than robust fiscal frameworks to fund SWM operations. This study analyzed the impact of municipal tax regulation and sustainable taxation practices—specifically differentiated and progressive tariffs—on the income generation capacity of 240 Mexican municipalities. Data from municipal financial reports was extracted from official government websites and analyzed using statistical methods in R, including variance analysis and t-tests. The results demonstrate that municipalities with higher levels of tax regulation achieve significantly greater per capita income from SWM services compared to those with minimal or no regulation. Differentiated tariffs based on socioeconomic status were found to be particularly effective, generating income levels 3.29 times higher than uniform tariffs. In contrast, progressive tariffs based on waste quantity did not significantly enhance revenue generation. These findings challenge assumptions about the universal applicability of progressive tariffs and underscore the importance of tailoring fiscal strategies to local economic and social contexts. This study highlights the need for robust tax regulation and strategic fiscal policies to strengthen municipal revenue streams and improve SWM service delivery. It provides valuable information about the impact of tax regulation quality and sustainable fiscal practices on municipal income generation, offering insights for policymakers aiming to achieve fiscal sustainability and equitable access to SWM services.

Introduction

Municipal solid waste management: Obstacles and responsibilities

In accordance with the Political Constitution of the United Mexican States (Constitución Política de Los Estados Unidos Mexicanos [CPEUM], 2023), the General Law on Ecological Balance and Environmental Protection (Ley General Del Equilibrio Ecológico y La Protección al Ambiente [LGEEPA], 2015), as well as the General Law for the Prevention and Integral Management of Waste (Ley General Para La

Prevención y Gestión Integral de Los Residuos [LGPGIR], 2014), Mexican municipalities hold the responsibility for the provision of public services related to the cleaning, collection, transportation, treatment, and final disposal of waste. Furthermore, they are tasked with implementing regulations aimed at preventing and controlling the environmental impacts arising from these services. They are also authorized to formulate municipal programs for the comprehensive management of urban solid waste, enact relevant regulations, and impose charges for the services provided.

Based on the findings from the National Censuses of Municipal

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Governments and Territorial Demarcations in Mexico City for the years 2021¹ and 2023,² a mere 34 out of 2475 (as of 2022) municipal entities and territorial demarcations (approximately 1.4 %) manage to channel a portion of their urban solid waste to treatment facilities (National Institute of Statistics and Geography, 2023). In contrast, 1188 (48 %) handle waste disposal at final sites lacking waterproofing systems, and 1429 (58 %) deposit waste in locations where it remains uncovered and uncompacted. Furthermore, only 53 municipalities (2 %) have established a Municipal Program for the Prevention and Comprehensive Management of Urban Solid Waste (INEGI, 2023). This scenario is in part due to the substantial financial burdens associated with waste management operations and the inadequate or non-existent mechanisms for collecting funds to support these services.

Legal framework of taxation and income sources for Mexican municipalities

In accordance with article 115 of the Political Constitution of the United Mexican States, municipalities are granted the autonomy to administer their treasuries. Mexican municipal treasury is composed of various sources of income, including federal or state transfers and municipalities own revenues (Consejo Nacional de Armonización Contable [CONAC], 2018; Constitución Política de Los Estados Unidos Mexicanos [CPEUM], 2023). While the income derived from federal and state transfers is contingent upon factors beyond municipal control, municipalities' own revenue may be enhanced through the training of officials and the effective regulation of municipal treasury (Frisby Morales et al., 2010; Martínez Sánchez and Cano Moreno, 2022; Villalobos López, 2023).

Municipal treasury regulations, in accordance with the laws of the respective state, typically consist of statutes or codes that establish the legal framework for the identification of revenue sources and the determination of fees, rates, and tariffs applicable to each income category, in addition to the total amount anticipated to be collected for each (México - Congreso de la Unión, 2022, 2023, 2024). These documents are required to delineate which municipal public services will incur charges, the individuals subject to such payments, the tariff structure, the basis for calculation and payment frequency; while adhering to the principles of legality, equity, proportionality, and allocation to public spending (González García, 2023). As stated by Guillerme Peón and Vargas Casimiro (2017) and Caldera Sánchez (2013), revenue collection efficiency among Mexico's states varies significantly, with Mexico City being the best-performing state. This could be related to the revenue collection efficiency of its municipalities.

The task of characterizing or defining levels of environmental regulation has been undertaken by various institutions, as evidenced by the Environmental Democracy Index (Worker and de Silva, 2015), the Environmental Performance Index (Block et al., 2024), and the Environmental Rule of Law Global Report (United Nations Environment Programme [UNEP], 2019). Most of these initiatives have been developed at the national level, with subnational approaches being infrequent. Nevertheless, there is no documented evidence of indices focused on public solid waste management, particularly at the local level.

Charging for waste management public services as an economic instrument

Tax policies can mobilize resources by enhancing revenue collection mechanisms, mitigate inequalities, and concurrently advance sustainable consumption and production frameworks. It is important to highlight that, while fiscal policies are a key component in resource mobilization, their effectiveness depends on the efficiency of the collection mechanisms, transparency in tax management, and the

government's ability to implement measures that encourage tax compliance. According to (Prasetyanti et al., 2023), optimizing these mechanisms can have a significant impact on a country's ability to finance its development needs and public services.

In essence, the structure of these policies can shape individual behaviors and exert a positive influence on the fulfillment of the Sustainable Development Goals (SDGs) (UNDP, 2023). Within the waste management sector, sustainable practices that contribute to enduring sustainability encompass methods that sustain separate waste streams, incentivize the minimization of waste generation and promote a circular economy. The imposition of tariffs for waste management public services, as an economic instrument, is increasingly acknowledged as an effective approach to improve waste management performance and encourage sustainable practices among users (André and Cerdá, 2006; García López, 2013; Matheson, 2022a; Monteiro dos Santos and Soares de Castro, 2021; Puig Ventosa, 2002).

The adoption of progressive pricing mechanisms based on the quantity of waste produced aligns with the "polluter pays" principle. This principle asserts that waste producers should assume the financial responsibility for its management, thus incentivizing the reduction of waste and the adoption of adequate sorting practices (Beccarello and Di Foggia, 2022; Morlok et al., 2017a; Panayotou, 1995). Unit pricing schemes, such as Pay-As-You-Throw (PAYT), have proven effective in enhancing recycling rates and curbing overall waste generation by explicitly linking the cost of waste management to the volume of waste produced (Morlok et al., 2017a). Such strategies have been implemented in some places across the world, sometimes as a nationwide policy as in Ireland (Davies and O'Callaghan-Platt, 2008), and other times as a local effort as in the county of Aschaffenburg, Germany (Morlok et al., 2017b). In Aschaffenburg, Germany, waste management is centered on the PAYT system, which charges residents based on the weight or volume of their waste. This system, supported by advanced infrastructure and public awareness campaigns, ensures efficient waste separation and treatment. Residual waste is incinerated, organic waste is composted or anaerobically digested, and recyclables are processed. The program has achieved an 86 % recycling rate and significantly reduced final waste, demonstrating its environmental and economic effectiveness (Morlok et al., 2017b). In Ireland, a similar Pay-By-Use (PBU) system was introduced in 2005 to encourage waste reduction and recycling. Operated under a mix of public and private management, it employs a two-tier pricing model with fixed and variable charges. Despite challenges such as inconsistent regulation and cost disparities, the system has increased recycling and reduced landfill use (Davies and O'Callaghan-Platt, 2008).

In Mexico, tariffs for waste management services are rarely implemented. In Villaflores, Chiapas, operational inefficiencies persist without user fees, despite challenges like limited collection coverage and deficiencies in disposal practices (Araiza Aguilar et al., 2017). In Guadalajara, Jalisco, waste management is hindered by administrative inefficiencies, and there is no evidence of specific tariffs for collection or recycling (Toledo Cervantes and Quintero Castellanos, 2022). A broader study of 66 municipalities confirms that most lack official tariffs, complicating cost recovery and sustainability (Turcott Cervantes et al., 2021).

Moreover, the implementation of tiered tariff packages for waste management services, as posited by Awunyo-Vitor et al (2013) and Vieira and Matheus (2018), can effectively mitigate the financial inequalities observed among households. By devising tariffs tailored to low, medium, and high-income households, municipalities can guarantee the coverage of waste management costs while simultaneously fostering incentives for waste minimization. Specifically, households within lower-income brackets may necessitate subsidized tariffs or tiered pricing frameworks to ensure the accessibility and fairness of waste management services (Elrabaya and Marchenko, 2022; Elrabaya, 2022). This necessity is substantiated by the observation that fixed tariffs frequently fail to incentivize waste reduction among lower-

¹ <https://www.inegi.org.mx/programas/cngmd/2021/>.

² <https://www.inegi.org.mx/programas/cngmd/2023/>.

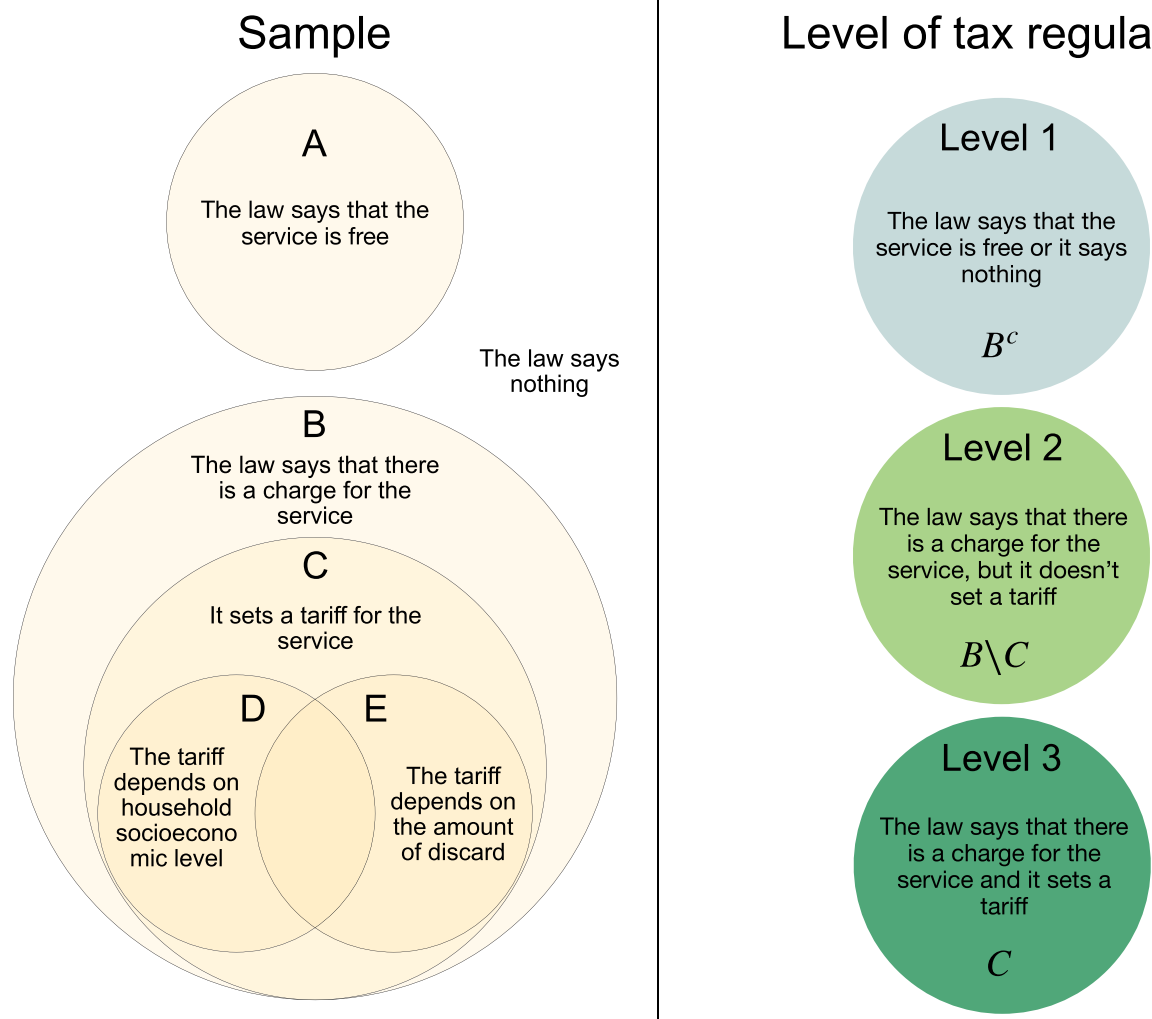


Fig. 1. Sample subsets and levels of tax regulation.

income households, as they may not correspond with the actual volume of waste produced (Di Foggia and Beccarello, 2020; Elrabaya and Marchenko, 2022).

Nevertheless, the deployment of these instruments encounters significant challenges, including the absence of specific regulations, inconsistency of initiatives, insufficient institutional capacity, lack of political support, inadequate legal frameworks, data scarcity, financial constraints and limited human resources (Besen and Fracalanza, 2016; de Lorena Diniz Chaves et al., 2014; Massoud et al., 2021; Matheson, 2022b; Monteiro dos Santos and Soares de Castro, 2021). Modifications in regulatory frameworks and the integration of economic incentives are crucial for the optimization of waste management services and the reduction of greenhouse gas emissions in Latin American countries (Rondinel-Oviedo, 2021; Sun and Asari, 2023; Yegorov et al., 2022).

Objectives

The objectives of this study are to examine the extent to which three different variables affect the municipal ability to generate income for the provision of public SWM services. These are: 1) quality of municipal tax regulation affects the capability to generate income for the provision of public waste management services, 2) differentiated tariffs based on socioeconomic status, and 3) progressive tariffs.

We therefore hypothesise:

- h1: The mean income per capita in municipalities in tax regulation level 2 is superior to that in municipalities in tax regulation level 1.
- h2: The mean income per capita in municipalities in tax regulation level 3 exceeds that in municipalities in tax regulation level 1.
- h3: The mean income per capita in municipalities in tax regulation level 3 surpasses that in municipalities in tax regulation level 2.
- h4: The mean income per capita in municipalities with a differentiated tariff structure is higher than that in municipalities with a non-differentiated tariff structure.
- h5: The mean income per capita in municipalities with a progressive tariff structure is greater than that in municipalities with a non-progressive tariff structure.

Methods

Sampling frame

Historical data sourced from the Public Finance Statistical Information Program (INEGI, 2024) encompassing the self-reported and non-audited financial figures of revenues and expenditures pertinent to federal entities, municipalities, and local governments nationwide was downloaded from the official website.³ Subsequently, the data was loaded

³ <https://www.inegi.org.mx/programas/finanzas/>.

Table 1
Sustainable tax practices variables and codes.

Variable	Value 1 (Yes)	Value 0 (No)
Differentiated	The fee varies based on the user's socioeconomic status.	The fee is the same for all users, regardless of their socioeconomic status.
Progressive	The fee depends on the amount of waste generated.	The fee is fixed and does not depend on the amount of waste generated.
Separation	There is an incentive or discount for users who separate waste at the household level.	There are no incentives for separating waste at the household level.

into RStudio, where it underwent a filtering process to exclusively retain the revenue category specified as “Cleaning and garbage collection services”; which includes income from municipal waste management public services (streets cleaning, collection, transportation, treatment, transfer and final deposit) in Mexico, as paid by citizens, but excludes income related to productive activities and resources or assets exploitation. Ultimately, data from the fiscal year 2022 was selected owing to its status as the most recent year with definitive figures available at the time of consultation. Although data from previous fiscal years is also available, it is not under the scope of this study and including it would require an impossible amount of work given the resources available. This process resulted in a dataset comprising 660 municipalities reporting revenues from these services. The local governments of Mexico City were excluded from this process, as the information program omits their revenue data (INEGI, 2016).

Variables definition

Level of municipal tax regulation on household solid waste management

The municipal level of tax regulation regarding household solid waste (LTR) is a categorical variable that organizes municipalities

reporting revenues related to solid waste management into three different levels. Level 1 encompasses municipalities that omit any reference to public waste management services within their financial or income regulations or codes. Level 2 comprises municipalities that impose a charge on household users who qualify for solid waste management services. Finally, Level 3 involves those that set a specific tariff imposed upon these users. Every municipality falls into one of such three levels. This categorization is depicted in Fig. 1.

Sustainable tax practices

Sustainable tax practices are defined as tax strategies that can effectively contribute to achieving specific sustainable development targets (UNDP, 2023). In the context of SWM, they are those that in addition to fortifying municipal financial structures, facilitate the implementation of public policies designed to reduce solid waste and its environmental impacts reduction within municipalities, promoting circular economy practices. To evaluate the implementation of such practices, two dichotomous variables were established, coded as 0 (No) and 1 (Yes). The variable “Differentiated” examines whether the prescribed tariff fluctuates based on the socioeconomic status of the service user, “Progressive” considers whether the tariff is contingent upon the quantity of solid waste generated, and “Separation” evaluates whether the tariff or applied discounts incentivize waste separation at the household level, as summarized in Table 1.

Sample size

The categorical variables considered to determine the sample size were level of tax regulation, Differentiated, Progressive and Separation. Since there is no evidence of the proportion of each category for any of such variables, the formula to determine a sample size adjusted for finite population (Equation 1), as suggested by Martínez Bencardino (2012),

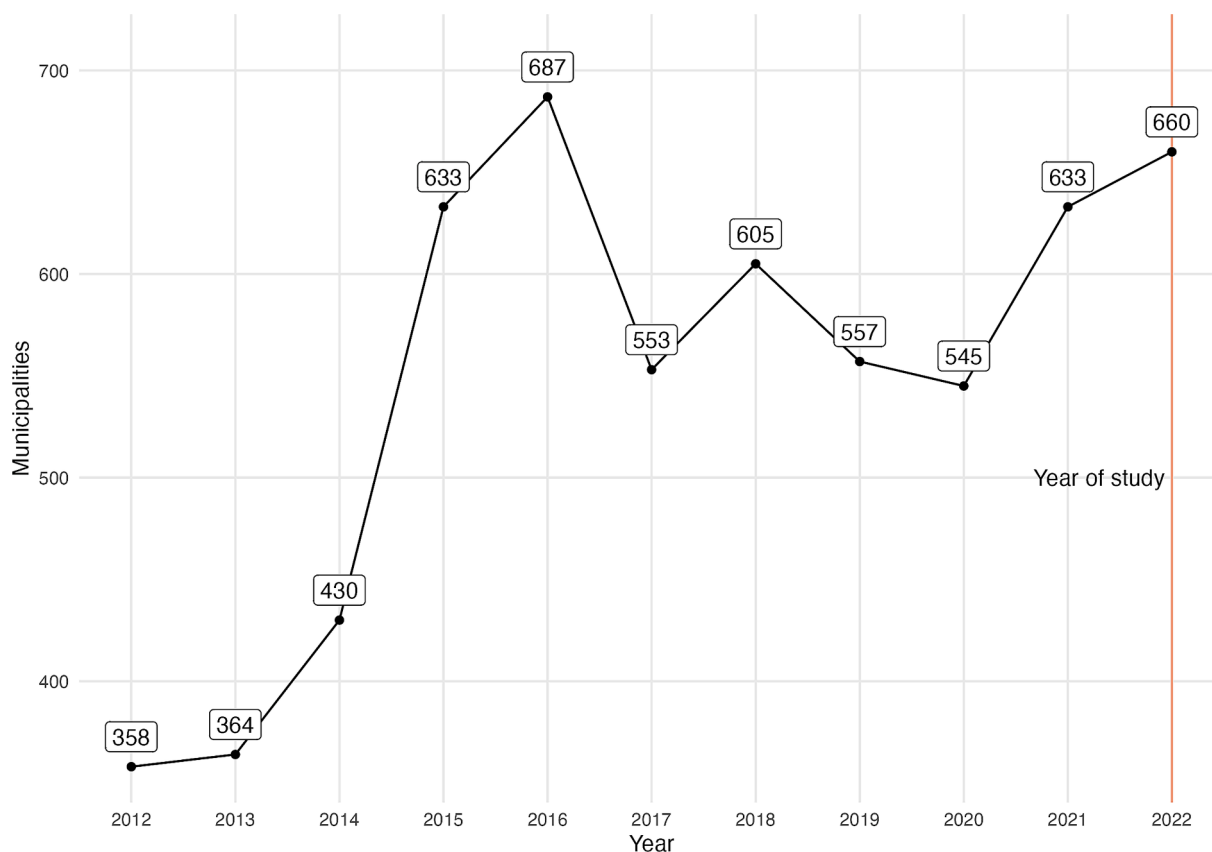


Fig. 2. Number of municipalities that reported income from the provision of solid waste management services over time (2012–2022).

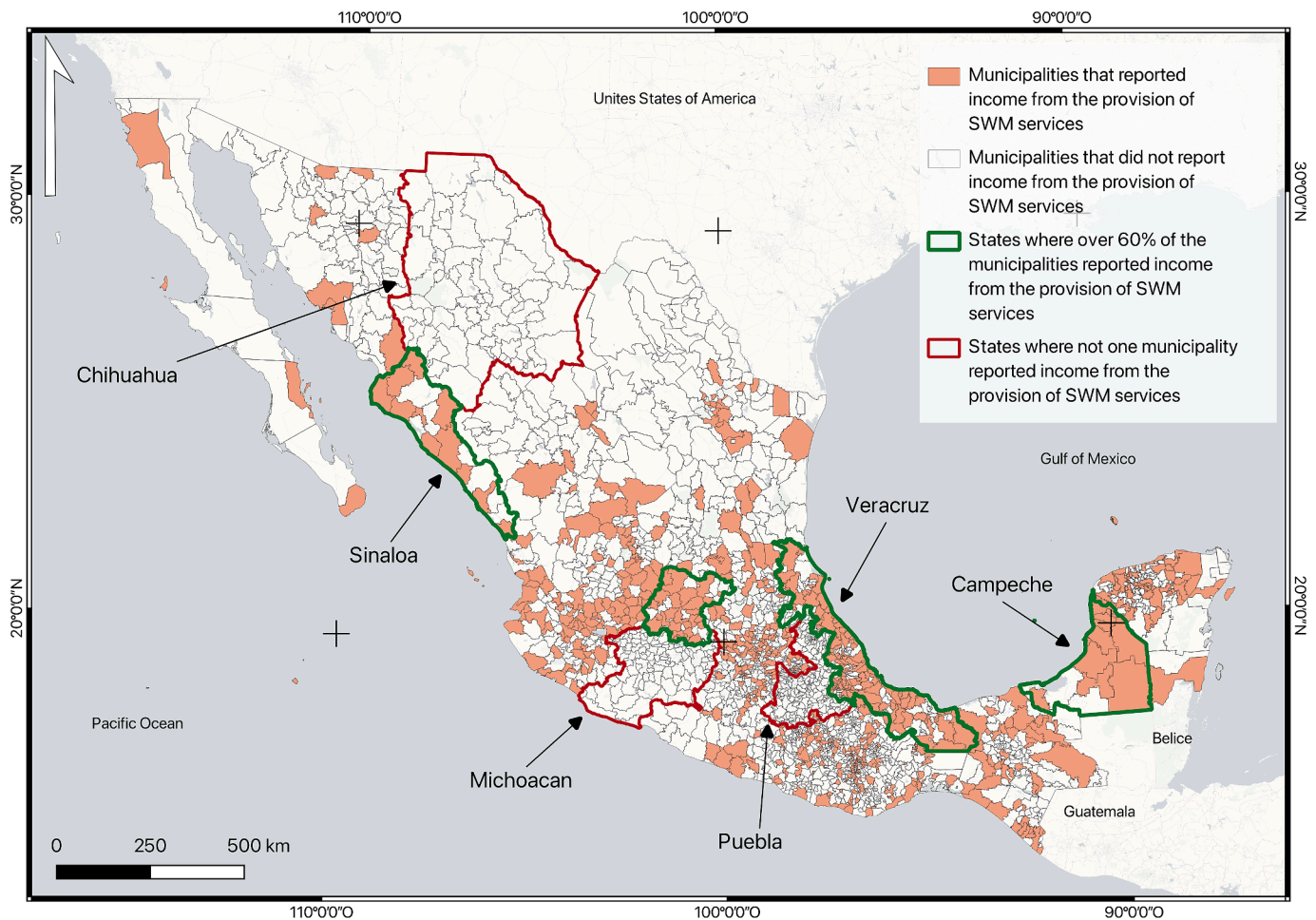


Fig. 3. Municipalities that reported income from the provision of solid waste management services in 2022.

was employed with an estimated proportion of 0.5, adhering to a 95 % confidence level and a maximum allowable margin of error of 0.05.

$$n = \frac{Z^2 p(1-p)}{e^2 + \frac{Z^2 p(1-p)}{N}} \quad (1)$$

where:

n : sample size.

$Z = 1.96$: this represents the critical value in the normal distribution that aligns with a 95 % confidence level.

p : the estimated proportion for the variable is 0.5.

e : 5 % error.

N : 660: the number of municipalities that generated revenues from “Cleaning and garbage collection services” during the 2022 fiscal year.

The determined sample size amounted to 244.

Sampling

A sample of 244 municipalities out of the 660 (N = number of municipalities that generated revenues from “Cleaning and garbage collection services” during the 2022 fiscal year.) was selected through a random process without replacement. A comprehensive search was conducted on the websites of state and local governments, as well as through commercial search engines, to locate the income laws, tax codes, and income budgets of the municipalities that make up the sample. These documents were then analyzed to determine whether the

municipal legal framework explicitly specified whether street cleaning, collection, treatment, transport, and final disposal of household solid waste services were free of charge or subject to a tariff.

Upon identifying the existence of a tariff for these services, the same legal documents were once again read to ascertain whether a tariff structure had been formalized, whether the tariffs *were differentiated and progressive*, whether provisions for the frequency and method of payment were delineated, and whether incentives for waste *segregation* at the source were in place. Furthermore, an examination of such legal framework was undertaken to determine if distinct tariff structures were prescribed for non-household users.

After this analysis, municipal income legislation and budgets were examined to determine the estimated income that the municipality anticipated acquiring for these services in the fiscal year.

Statistical analysis

Descriptive analysis of variables

A descriptive analysis was conducted for the study variables. Proportions for each level were calculated for the LTR variable. Subsequently, the same was done for the variables Differentiated and Progressive for the subset of the sample that makes up level 3.

Municipal per capita income

The mean of PCInc for the whole population is 18.52499, as shown in the [supplementary material](#). This variable is defined by equation 2.

$$PcInc = (100 * SWMInc) / (TotPop * PSCov) \quad (2)$$

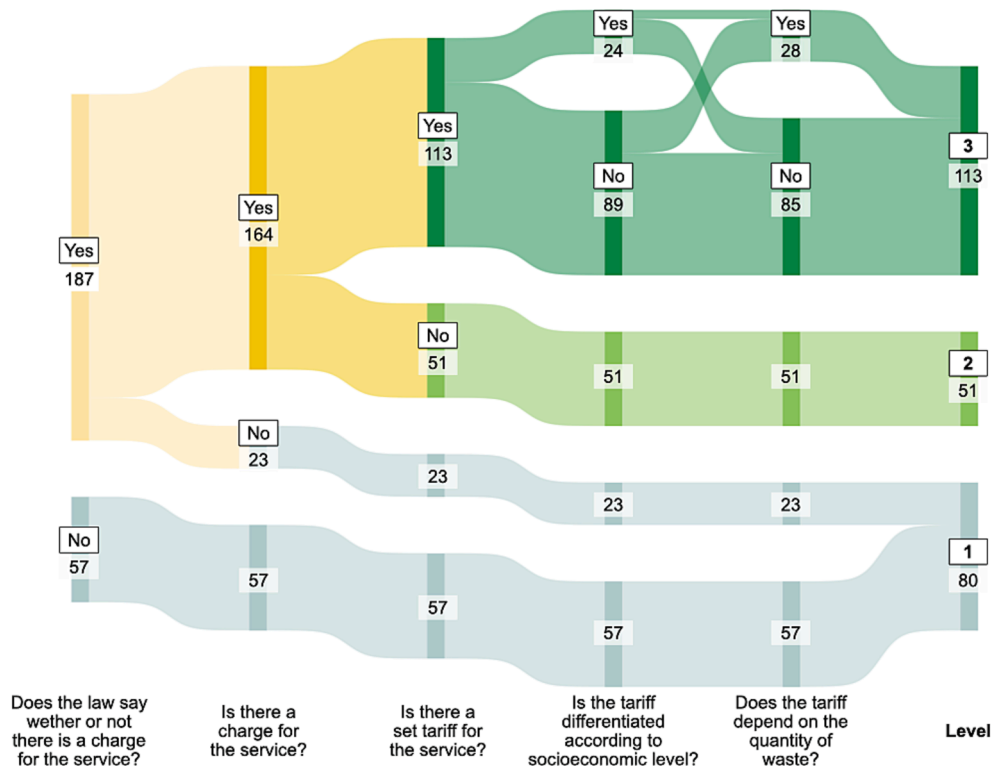


Fig. 4. Levels of tax regulation. Note: numbers on the graph represent municipalities of the sample; whereas percentages in the text above represent the population parameters that resulted from statistical inference.

where:

PcInc: annual municipal income derived from waste management activities per capita, considering the population who benefits from solid waste collection services.

SWMInc: annual municipal income derived from waste management activities services.

TotPop: municipality's total population.

PSCov: coverage of public waste management services (%).

Logarithmic transformation

A logarithmic transformation was implemented on the *PcInc* variable, as suggested by Lee (2020), resulting in the generation of a new variable designated as *logPcInc*, as defined by Equation 3.

$$\log PI = \log PcInc \quad (3)$$

where,

PcInc: annual municipal income derived from waste management activities per capita, considering the population who benefits from solid waste collection services.

logIngPcC: natural logarithm of *PcInc*.

Analysis of variance

In order to ascertain whether the LTR variable exerts an influence on *PcInc*, histograms and quantile–quantile plots were constructed, and the Shapiro-Wilk test and the Anderson-Darling test were applied to evaluate the normality of residuals; and a plot of residuals against fitted values was produced in conjunction with Bartlett's test to examine the homogeneity of variances, as both are essential conditions to perform an analysis of variance. Then such analysis of variance accompanied by Tukey's test, was initially employed to evaluate the presence of significant differences among the means of the *logPI* variable across the three distinct levels of LTR. Thereafter, the magnitude of these differences was

quantified utilizing Equation 4, as suggested by (Lee, 2020).

$$\text{Factor} = \exp(\log IP_n) / \exp(\log IP_m) \quad (4)$$

where,

$\log PI_n$: natural logarithm of *PcInc* for level *n*.

$\log PI_m$: natural logarithm of *PcInc* for level *m*.

Student's *t* test

To determine if the Differentiated and Progressive variables have any impact on the *IngPcC* variable, histograms and quantile–quantile plots were constructed, the Shapiro-Wilk test and the Anderson-Darling test were applied to evaluate the normality of residuals; and a plot of residuals against fitted values was produced in conjunction with Levene's test and Brown-Forsythe's test to examine the homogeneity of variances, as both are essential conditions to perform a Student's *t*-test analysis. Then such Student's *t*-test was applied and the extent of significant differences in group means was measured using Equation 4.

Results

Municipal income from solid waste management services

Fig. 2 illustrates a pronounced escalation in the number of municipalities reporting income derived from the provision of solid waste management services between 2012 and 2016, increasing from 358 to 687. Subsequently, from 2017 onwards, there is an observable decline in the number of municipalities reporting such income, which then remains relatively stable until 2022.

Fig. 3 illustrates the disparate spatial distribution of municipalities that reported income sourced from the provision of solid waste management services across the federal entities of Mexico. Notably, municipalities in Puebla, Michoacán, and Chihuahua did not report any income from these services, whereas over 60 % of the municipalities in

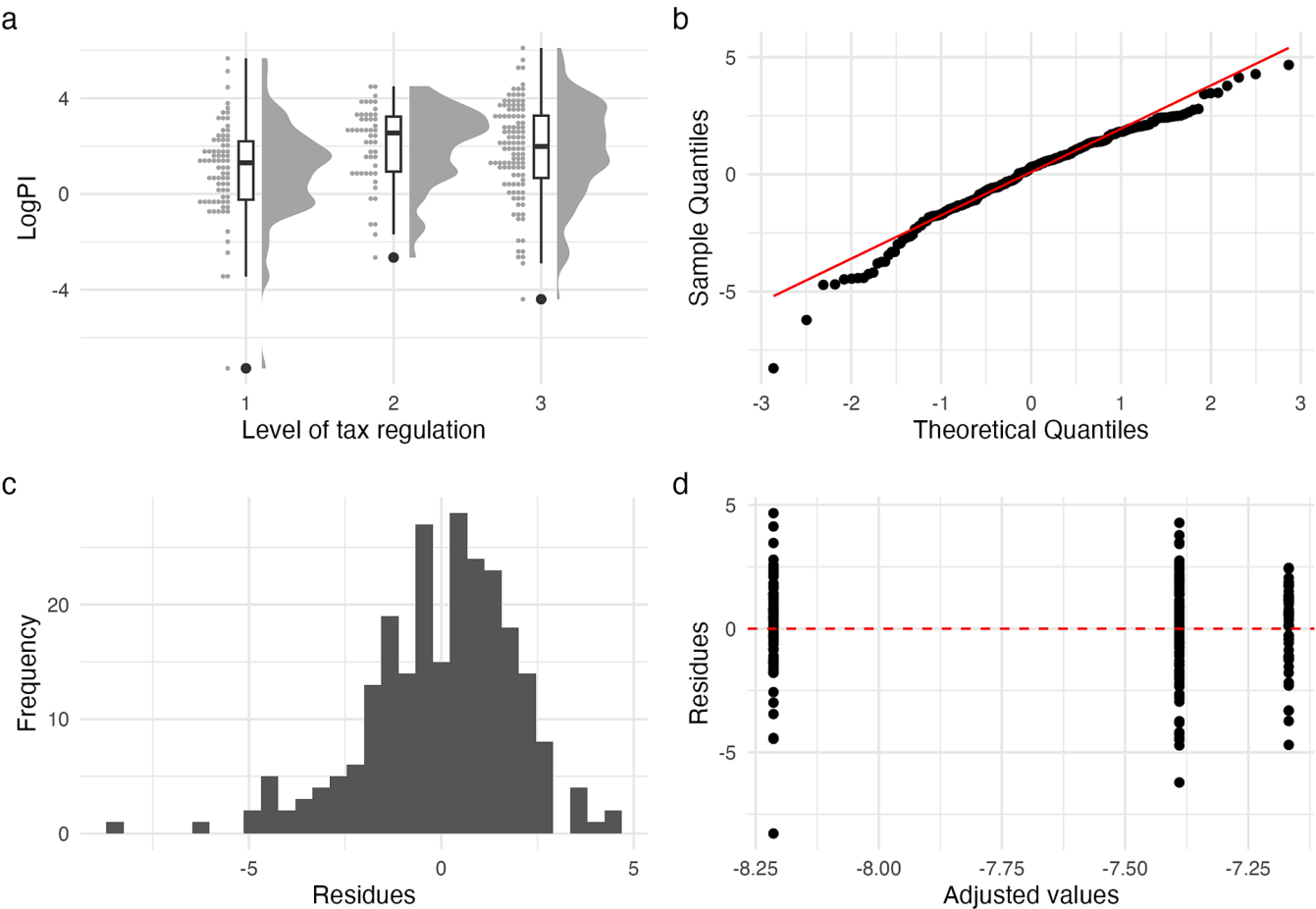


Fig. 5. Statistical tests for LogIP Level: a) Boxplot, b) QQ plot, c) residues histogram and d) adjusted values versus residues plot.

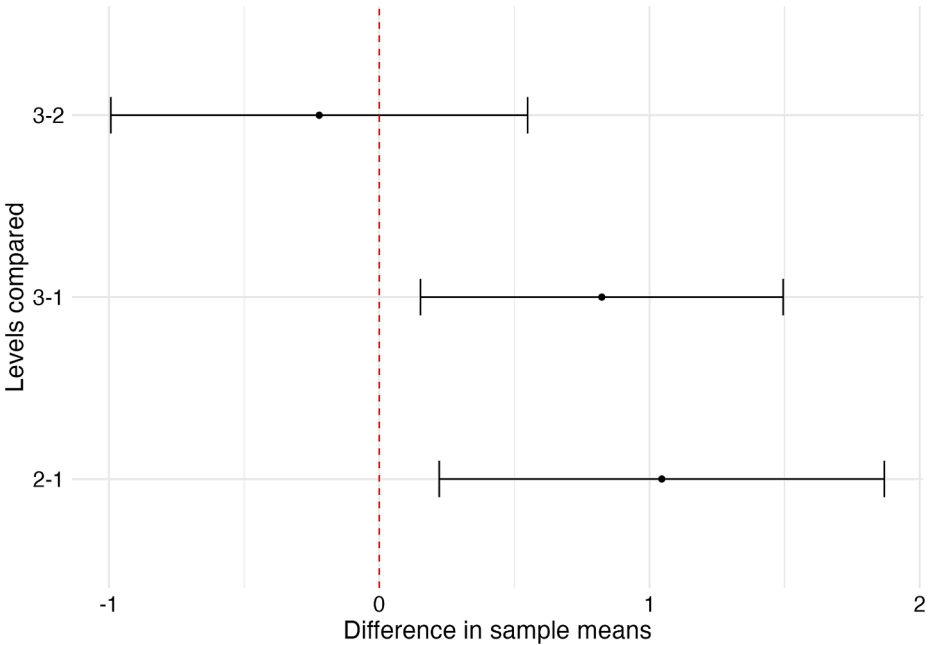


Fig. 6. Differences between sample means.

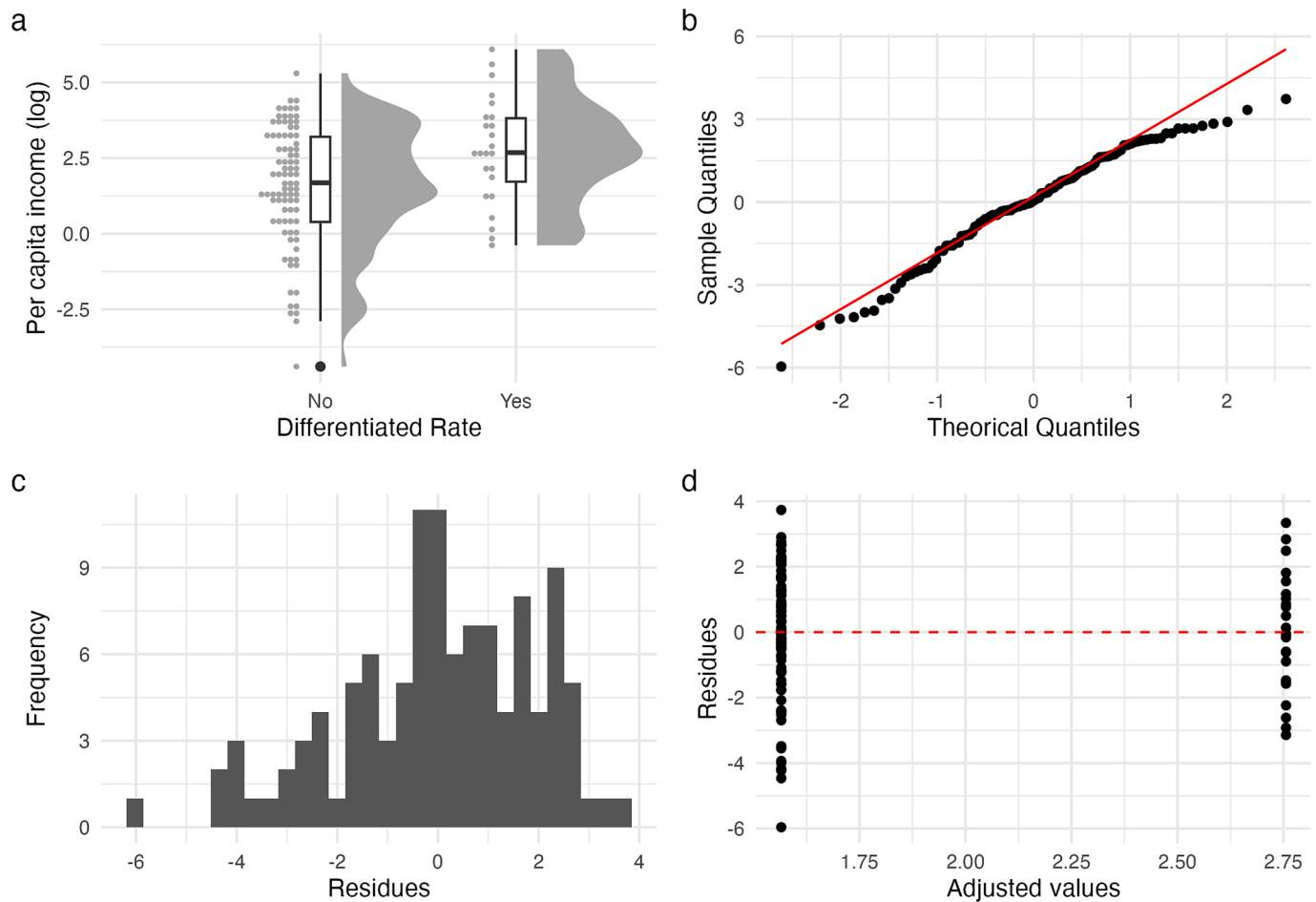


Fig. 7. Statistical tests for LogIP Differentiated: a) Boxplot, b) QQ plot, c) residues histogram and d) adjusted values versus residues plot.

Veracruz, Sinaloa, Guanajuato, and Campeche reported such income.

Municipal tax regulation on household solid waste management

As illustrated in Fig. 4, most municipal regulations (76.6 % $\pm$ 5.31 %) explicitly indicate whether charges are imposed for waste management services; a substantial proportion of these (87.6 % $\pm$ 4.21 %) indeed implement such charges; and 68.9 % ($\pm$ 6.29 %) of these regulations establish the specific tariff for such payments. Consequently, the first level of tax regulation encompasses 80 (32.79 % $\pm$ 5.27 %) municipalities, while the second and third levels comprise 51 (20.90 % $\pm$ 4.78 %) and 113 (46.31 % $\pm$ 6.31 %) municipalities, respectively. Moreover, within the 113 municipalities categorized under level 3, a mere 24 (21.24 % $\pm$ 7.34 %) implement differentiated tariffs contingent on household socioeconomic status, and 28 (24.78 % $\pm$ 7.82 %) municipalities implement progressive tariffs based on the quantity of waste produced. Only 5 (4.42 % $\pm$ 3.76 %) of them implement both differentiated and progressive tariffs.

The impact of tax regulation level on the income generation capacity of municipalities derived from solid waste management public services.

Fig. 5a illustrates a boxplot of the per capita income from waste management services across the three tax regulation levels, where the 80 municipalities in Level 1 exhibit the lowest mean income (mean = 13.2, standard deviation = 38.8) from waste management services provision, while those in Levels 2 (mean = 18.1, standard deviation = 20.8) and 3 (mean = 26.1, standard deviation = 56.2) show higher, though similar, distributions. The Shapiro-Wilk test results ($W = 0.96926$, $p\text{-value} =$

$4.863\text{e-}05$) and the Anderson-Darling test results ($A = 1.6279$, $p\text{-value} = 0.0003423$) suggest a deviation from normal distribution in the model residuals; however, this is contradicted by the evidence presented in Fig. 5b, where the QQ plot shows the residuals largely aligning with the reference line, suggesting minimal deviations from normality. Fig. 5c further supports this with a histogram of the residuals, showing an approximately symmetrical distribution with slight skewness. In contrast, both Bartlett's test outcomes ($K\text{-squared} = 2.7258$, $df = 2$, $p\text{-value} = 0.2559$) and Fig. 5d, which displays the scatter plot of adjusted values versus residuals, collectively demonstrate that the variance across the three groups is homogenous, as no discernible patterns are observed.

As shown in Fig. 6, on the one hand, there is a significant difference between the means of levels 1 and 2 and levels 1 and 3. On the other hand, there is no such difference between levels 2 and 3. In this sense, the mean income per person with waste collection services in level 2 is 2.845092 times that of level 1, while the mean in level 3 is 2.278798 times higher than that of level 1.

The impact of sustainable tax practices on the income generation capacity of municipalities derived from solid waste management public services.

Differentiated tariff

Fig. 7a illustrates a boxplot that compares the PcInc between municipalities with and without differentiated tariffs based on socioeconomic status or housing type. The municipalities implementing differentiated tariffs exhibit significantly higher mean PcInc (mean = 57.3, $sd = 104$) compared to those without such a structure (mean = 17.6, $sd = 28.5$). Fig. 7b shows a QQ plot of residuals, indicating that

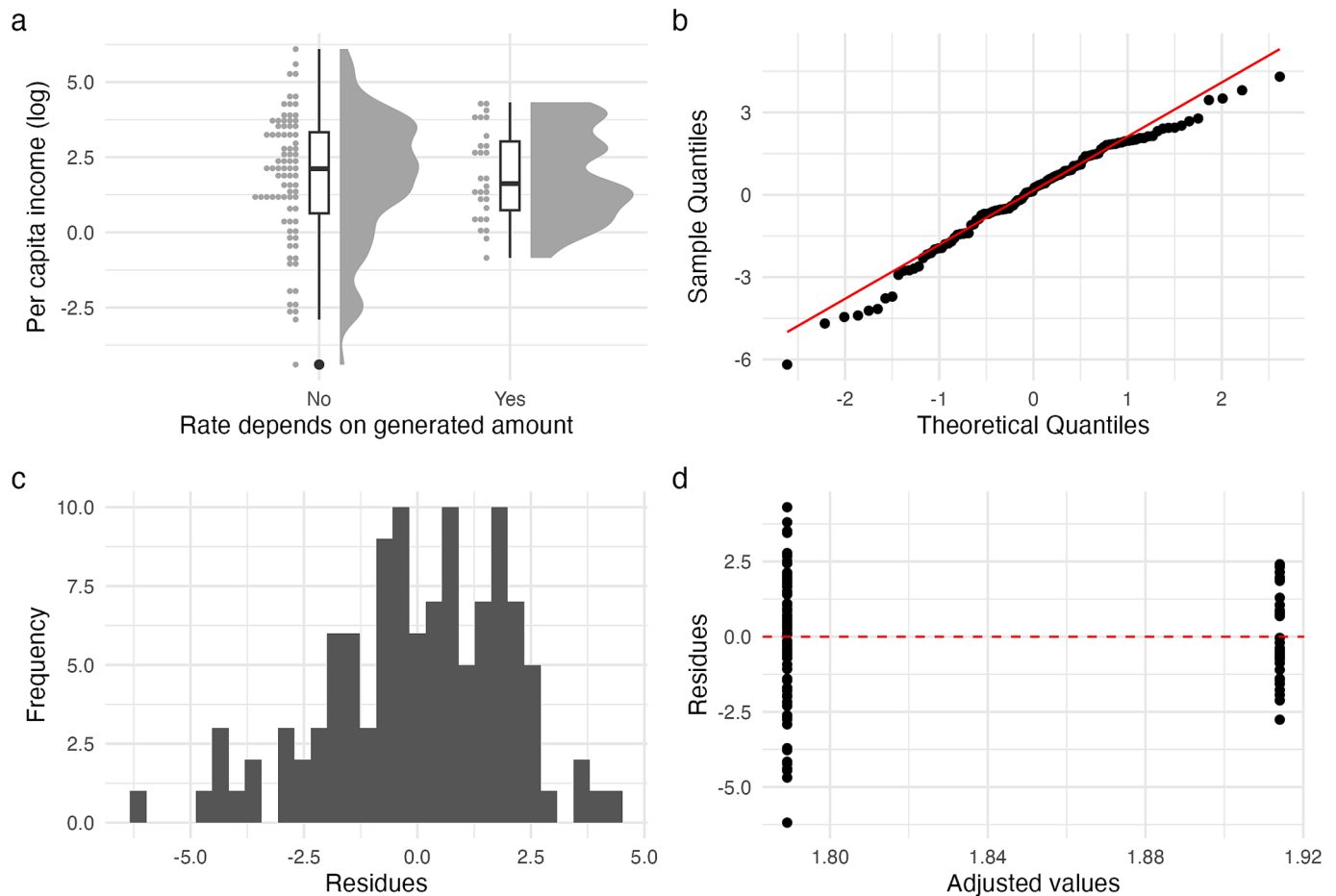


Fig. 8. Statistical tests for *LogIP Progressive*: a) Boxplot, b) QQ plot, c) residues histogram and d) adjusted values versus residues plot.

despite minor deviations, the data largely aligns with the reference line, suggesting that the assumption of normality is reasonable for this analysis. Fig. 7c further supports this conclusion by presenting a histogram of residuals, which demonstrates an approximately symmetrical distribution with minimal skewness. Fig. 7d provides a scatter plot of adjusted values versus residuals, which confirms the homogeneity of variances, as no discernible patterns are observed.

Considering that the Student's *t*-test exhibits robustness to deviations from normality of residuals in large sample sizes, the graphical analyses illustrated in Fig. 7 are deemed adequate to infer a normal distribution. However, it is noteworthy that both the Shapiro-Wilk test ($W = 0.97223$, $p\text{-value} = 0.01949$) and the Anderson-Darling test ($A = 0.78013$, $p\text{-value} = 0.04162$) reject the null hypothesis positing that the sample is derived from a normal distribution. Conversely, the outcomes of Levene's test ($DF = 1$, $F\text{ value} = 0.5427$, $\Pr(>F) = 0.4629$) and the Brown-Forsythe's test ($DF = 1$, $F\text{ value} = 0.5435$, $\Pr(>F) = 0.4626$) along with the graphical representation corroborate the similarity in variances between the two groups.

The outcome of the Student's *t*-test ($t = -2.6498$, $df = 110$, $p\text{-value} = 0.009239$) leads to a rejection of the null hypothesis, which posits no differences between sample means. This finding substantiates a statistically significant disparity between municipalities that impose a differentiated tariff based on socioeconomic level or housing type and those that do not. Consequently, the mean income per person utilizing waste collection services in municipalities with a differentiated tariff structure is 3.290013 times greater than that in municipalities with a uniform tariff structure.

Progressive tariff

The evaluation of progressive tariffs—where fees depend on the amount of waste generated—did not yield significant differences in income generation capacity, as illustrated in Fig. 8. Fig. 8a presents a boxplot comparing municipalities that implement progressive tariffs with those that do not. Unlike the results observed for differentiated tariffs, both groups exhibit similar *PcInc* distributions, with no clear advantage for municipalities adopting a progressive tariff structure. Eventhough, the Shapiro-Wilk test results ($W = 0.97624$, $p\text{-value} = 0.04309$) and the Anderson-Darling results ($A = 0.75418$, $p\text{-value} = 0.04828$) indicate a rejection of the null hypothesis, suggesting that the sample may not originate from a normal distribution, Fig. 8b (QQ plot) shows that while minor deviations from normality are present, the data largely align with the expected distribution. This conclusion is reinforced by the histogram of residuals in Fig. 8c, where the overall symmetry of the data indicates that the deviations are not severe. Additionally, the Levene test ($DF = 1$, $F\text{ value} = 2.3972$, $\Pr(>F) = 0.1244$) and the Brow-Forsythe's test ($DF = 1$, $F\text{ value} = 2.0735$, $\Pr(>F) = 0.1527$), alongside Fig. 8d (scatter plot of adjusted values versus residuals) confirm that variances between the two groups remain homogeneous, as evidenced by the absence of patterns in the residuals.

The outcomes of the Student's *t*-test ($t = -0.28449$, $df = 110$, $p\text{-value} = 0.7766$) validate these findings, as the $p\text{-value}$ exceeds the significance threshold, leading to the conclusion that the difference in mean *PcInc* between the two groups is not statistically significant.

Discussion

As anticipated, it was observed that municipalities categorized under

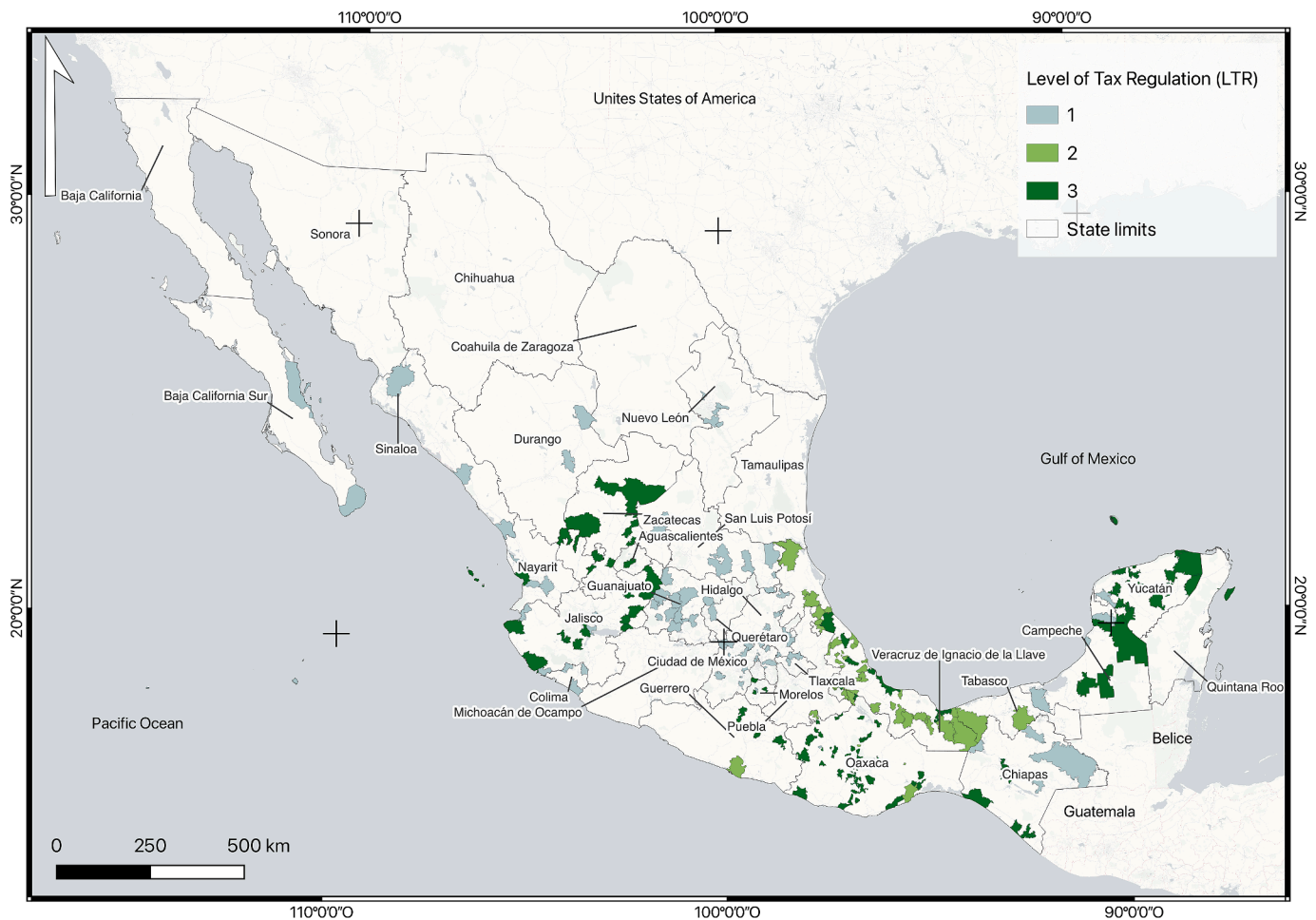


Fig. 9. Level of Tax Regulation and Mexican states.

higher levels of tax regulation (levels 2 and 3) exhibit substantially higher per capita revenue from waste management tariffs compared to those under the lowest level (level 1). Thus, we can say that level 2 represents the ideal standard for solid waste management tax regulation related. This finding aligns with the literature, which highlights the importance of appropriate tax policies and enforcement mechanisms for improving municipal solid waste management and associated revenue generation (André and Cerdá, 2006; García López, 2013; Kyere et al., 2019; Matheson, 2022a; Monteiro dos Santos and Soares de Castro, 2021; Puig Ventosa, 2002). Furthermore, this situation seems to be partially determined by state public policies, as it was observed that municipal laws and codes adhere to formats developed by state governments, and the proportions of municipalities reporting revenue from public waste management services vary widely between states. Fig. 9 shows that every municipality in the states of Baja California Sur, Sinaloa, Durango, Nuevo León, San Luis Potosí, Guanajuato, Querétaro, Hidalgo, Estado de México, Colima and Tlaxcala have a LTR 1; while both municipalities in Tabasco and most of those in Veracruz have a LTR 2; finally, all of those in Zacatecas, Aguascalientes and most of the ones in Oaxaca, Jalisco, Campeche and Yucatán have a LTR 3.

Moreover, the analysis on the impact of differentiated tariffs based on socioeconomic status or housing type reveals that this practice is associated with significantly higher income per person utilizing waste collection services. This corroborates previous studies that emphasize the significance of implementing progressive or differential pricing schemes to enhance the financial sustainability of municipal solid waste management (Di Foggia and Beccarello, 2020; Kinemo, 2019; Morlok et al., 2017a; Torrente-Velásquez et al., 2020). In contrast, the study did

not find a statistically significant relationship between the implementation of progressive tariffs based on waste quantity and the income generation capacity of municipalities.

However, this study has some limitations. The reliance on self-reported municipal data on income and SWM services coverage may not always be certain. There is also not available data about citizen satisfaction at a local scale and municipal expenditure on solid waste management and tax collection, which could help to better understand the financial and operational efficiency of municipal solid waste management systems. Additionally, the study did not assess whether progressive tariffs yield indirect benefits, such as encouraging waste reduction or segregation.

Conclusion

The findings of this study demonstrates that the level of tax regulation on solid waste management services significantly influences the income generation capacity of municipalities. These findings have important implications for policy makers and municipal authorities responsible for the management of solid waste. First, they underscore the need to strengthen tax regulation and enforcement mechanisms to improve the financial viability of solid waste management services. Second, the results show that the adoption of differentiated tariff structures, tailored to the socioeconomic conditions of the population, is a more effective strategy to improve revenue generation compared to progressive tariffs based on waste quantity. However, the absence of significant gains from progressive tariffs indicates that more research is needed to refine and better implement these mechanisms.

In general, this work contributes to understanding the financial dynamics of waste management services in Mexican municipalities and provides valuable information for policymakers looking to improve the fiscal autonomy of local governments through sustainable and equitable waste management practices. Moreover, this study aligns with the global trend, spearheaded by the United Nations, towards adopting sustainable fiscal practices. By promoting differentiated and progressive tariff structures, it supports the achievement of the Sustainable Development Goals (SDGs), particularly those related to environmental sustainability, reducing inequalities, and fostering responsible consumption and production patterns (SDG 10, SDG 11, and SDG 12).

Future research

Future research could investigate in depth the factors that influence the effectiveness of various tax policy approaches in the solid waste management domain, including an examination of contextual variables, institutional capacities, and community participation. It is also important to examine whether progressive tariffs have other benefits, such as incentivizing waste reduction, which escape the scope of this study.

Future research should delve into the factors influencing the effectiveness of various tax policy approaches in the solid waste management (SWM) domain, including contextual variables, institutional capacities, community participation, service quality, citizen satisfaction, financial efficiency, and environmental outcomes. Additionally, it is crucial to explore whether progressive tariffs provide benefits beyond income generation, such as incentivizing waste reduction and segregation, to gain a more comprehensive understanding of their role in promoting sustainable and equitable SWM systems.

CRediT authorship contribution statement

Sergio López-Olvera: Writing – review & editing, Writing – original draft, Visualization, Supervision, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Juan Carlos Olivo-Escudero:** Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Conceptualization. **Mauricio Hernández-Marroquín:** Writing – review & editing, Validation, Methodology, Formal analysis. **Fausto Rafael López-Olvera:** Writing – original draft, Investigation. **Yessica Lourdes Vallejo-Rivera:** Investigation. **Marco Antonio Rodríguez-Gómez:** Investigation. **Pedro Hipólito Rodríguez-Herrero:** Writing – review & editing, Project administration, Funding acquisition.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.wmb.2025.01.009>.

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